MICROBIAL DRUG RESISTANCE
Volume 31, Number 4, 2025

DOI: 10.1089/mdr.2024.0215

Minocycline and Omadacycline Resistance Among Carbapenem-Resistant Gram-Negative Bacteria: Antimicrobial Susceptibility Testing and Molecular Characterization

Richa Sinha,[1] Jyoti Jyoti,[1] Ashutosh Pathak,[1] Chinmoy Sahu,[2] Prabhaker Mishra,[3] Rungmei S.K. Marak,[1]
and Ujjala Ghoshal[4]

Increasing prevalence of multidrug-resistant infections has rendered the healthcare systems ineffective in managing infectious diseases. Drugs of "last resort" like carbapenems and polymyxins are becoming less effective in the management of antibiotic-resistant Gram-negative bacterial infections, leaving the clinicians with limited choices. Evaluation of the efficacy of other available broad-spectrum antibiotics (belonging to a different class) is warranted as a treatment alternative. The current study was undertaken to evaluate the *in vitro* antibacterial activity of minocycline and a new drug, omadacycline among carbapenem-resistant Gram-negative bacteria (GNB), isolated from clinical samples (pus and sputum) and to genotypically analyze them. A prospective cross-sectional study was conducted in a 3,200-bedded tertiary care medical center, located in Lucknow in the northern part of India. All the clinical isolates recovered from pus and sputum samples of patients admitted in intensive care units were processed according to the standard protocols. Identification and antibiotic susceptibility testing were performed, and carbapenem-resistant Gram-negative bacteria (CRGNB) showing resistance to minocycline were included in the study. Molecular screening of β-lactamase and tetracycline resistance genes was done by the conventional polymerase chain reaction method. Minimum inhibitory concentration analysis was performed using the broth microdilution technique. Among 700 CRGNB, 15.29% ($n = 107/700$) were minocycline resistant by disk diffusion method. Genetic analysis demonstrated the presence of tetracycline-resistant genes in about one-third isolates, among which the *tet*(B) gene was present in 41.12% ($n = 44/107$). Upon broth microdilution analysis, the overall minimum inhibitory concentration for minocycline was raised, wherein 4.76% ($n = 5/107$) of our clinical Gram-negative isolates were inhibited at ≤8 mg/L and 15.23% ($n = 28/107$) were inhibited at ≤16 mg/L. Omadacycline was able to inhibit 13.08% ($n = 14/107$) of the minocycline-resistant isolates at ≤4 mg/L (susceptible breakpoint for *Enterobacterales*). Based on the cut-off value proposed, 15.09% ($n = 16/107$) isolates resistant to minocycline were inhibited by omadacycline. High prevalence of multidrug-resistant bugs entails judicious use of minocycline and omadacycline. The presence of *tet* genes coexisting with bla_{NDM} and bla_{OXA} in our bacterial isolates shows that the resistance pattern in Gram-negative bacilli is regularly evolving, and a fully functional surveillance program across the health care system is needed to prevent the emergence and spread of antimicrobial resistance.

Keywords: carbapenem resistant, gram-negative bacteria, minocycline, omadacycline, MIC

[1]Sanjay Gandhi Post Graduate Institute of Medical Sciences, Microbiology, Lucknow, Uttar Pradesh, India.
[2]Sanjay Gandhi Post Graduate Institute of Medical Sciences, Lucknow, Uttar Pradesh, India.
[3]Sanjay Gandhi Post Graduate Institute of Medical Sciences, Department of Biostatistics and Health Informatics, Lucknow, Uttar Pradesh, India.
[4]AIIMS Kalyani, Microbiology, Kalyani, West Bengal, India.

Introduction

Infections caused by multidrug-resistant (MDR) Gram-negative bacteria (GNB) have increased exponentially in the recent past, affecting global health, food security, and development. It is well known that bacterial resistance is the result of extensive, unsupervised, and overuse of antibiotics. This has led to the rapid evolution of the bacterial genome, reducing the efficacy of most antimicrobials, thereby increasing selective pressure on antibiotic-resistant bacteria.[1] Carbapenems which were once considered the antibiotic of choice against multidrug-resistant (MDR) and extended spectrum β-lactamase (ESBL)-producing Enterobacteriaceae, have become less effective, thus limiting the treatment options.[2] Unprecedented use of colistin in human medicine, animal husbandry, aquaculture, and agriculture has squandered the efficacy of the drug and has promoted the emergence and dissemination of colistin resistance gene (*mcr*) among GNBs as well.[3] This entails the need for an antibiotic with a broader spectrum of activity. The development of a new class of antibiotic usually takes more than a decade. Minocycline, a second-generation tetracycline having a wide spectrum of activity, especially against drug-resistant microbes, has thus become an important treatment alternative. It is a bacteriostatic drug and can cross the blood–brain barrier easily. It inhibits the protein synthesis in bacteria by binding to the 30S ribosomal subunit.[4] It also has anti-inflammatory, antioxidant, antiapoptotic, and immunomodulatory effect. Compared to first-generation tetracycline, it has a better pharmacokinetic profile, longer half-life, and better absorption when administered orally.[5,6]

However, minocycline resistance among GNBs has been reported recently.[7,8] A variety of mechanisms have been proposed, including the efflux pump and modification or protection of the antibiotic target sites, few of which are linked to the *tet* genes.[9,10] As the resistance of microbes to the majority of the commonly prescribed antibiotics is increasing, it becomes prudent to initiate appropriate empirical therapy in time for the critically ill patients. Another drug that has recently gained popularity is omadacycline. Omadacycline, a semisynthetic derivative of minocycline, has recently been approved by the U.S. Food and Drug Administration (FDA) for the treatment of acute bacterial skin and skin structure infections and community-acquired bacterial pneumonia.[11–13] It has a broader spectrum of activity against Gram-positive and GNB, as well as atypical pathogens.[14] It is available in both intravenous and oral formulations. It is a potential last-resort antibiotic for difficult-to-treat bacterial infections, has once-daily oral dosing, and favorable pharmacokinetic properties.[15] These factors make omadacycline a promising alternative to existing tetracycline derivatives.[16] As the prevalence and antimicrobial resistance rates of microorganisms differ geographically, the current study was undertaken to evaluate the *in vitro* activity of minocycline and omadacycline in carbapenem-resistant GNB isolates. Another aim was the genotypic characterization of minocycline-resistant bacteria among carbapenem-resistant organisms.

Materials and Methods

A prospective cross-sectional study was conducted in the Department of Microbiology, Sanjay Gandhi Post Graduate Institute of Medical Science (SGPGIMS), a 1,600-bed tertiary care medical center located in Lucknow city of Uttar Pradesh between March 2022 and March 2024.

As the prevalence (p) of minocycline resistance among carbapenem-resistant GNB in our center is 16% (unpublished data), it was estimated that with a 1% margin of error and considering a design effect of 2, a total of 82 nonrepeat minocycline-resistant isolates among carbapenem-resistant clinical isolates would be required for the study, calculated by the formula of $n = Z^2p\ (1{-}p)/d^2$, where: "n" is the sample size, "Z" is the statistic corresponding to the level of confidence, "p" is expected prevalence (that can be obtained from the same studies or a pilot study) and "d" is the tolerated margin of error.

All the clinical isolates recovered from pus and sputum samples of patients admitted in intensive care units were primarily identified by standard conventional biochemical tests and further validated by the automated Matrix-Assisted Laser Desorption Ionization–Time of Flight (MALDI-TOF-MS) system (BioMerieux Inc., France). Antibiotic susceptibility testing by disk diffusion was performed according to the CLSI (Clinical and Laboratory Satndards Institute) guidelines, and Gram-negative carbapenem-resistant bacteria showing resistance to minocycline were included in the study. The microbes were stored in 20% glycerol medium at −80°C to be processed later for further evaluation of antibiogram and molecular analysis. At the time of further work-up (minimum inhibitory concentration [MIC] testing and molecular study), isolates were subcultured onto 5% sheep blood or chocolate agar. Genomic DNA of pathogens was extracted using the Qiagen Bacterial Genomic DNA isolation kit (Qiagen Inc., United States) according to the manufacturer's instructions. Molecular screening of β-lactamase genes ($_{bla}$NDM, $_{bla}$OXA 48) and tetracycline resistance genes (*tet*A, *tet*B, *tet*K, *tet*M, *tet*S) was done by the conventional PCR (polymerase chain reaction) method using specific primers (Table 1).[17–19] Interpretation of MIC was done using the microbroth dilution technique and was based on cut-offs proposed by CLSI for carbapenems and minocycline and the U.S. FDA for omadacycline.[20,21] Quality control was put up using ATCC isolates (American Type Culture Collection), that is, *Escherichia coli* strain ATCC 25922. Minocycline and omadacycline were procured from Cayman Chemical Company (Ann Arbor, Michigan, USA). The present study was approved by the institutional ethics committee of SGPGIMS (Reference No. PGI/DIR/RC/916/2021).

Result

A total of 700 carbapenem-resistant Gram-negative bacterial isolates (CRGNB) were screened by the disc diffusion method, and their antibiograms were documented. Among 700 isolates, 15.29% (n = 107/700) were found to be minocycline resistant. These minocycline-resistant bugs were further evaluated by the broth microdilution method as well. The distribution of various minocycline-resistant CRGNB was: 28.97% (n = 31/107) *Pseudomonas aeruginosa*, 25.23% (n = 27/107) *K. pneumoniae*, 14.95% (n = 16/107) *E. coli*, 13.08% (n = 14/107) *Acinetobacter baumannii*, 6.5% (n = 7/107) *Enterobacter spp.*, 5.6% (n = 6/107) *Citrobacter spp.*, 2.8% (n = 3/107) *Morganella spp.*, and 2.8% (n = 3/107) *Proteus spp.* (Table 2). Genetic characterization of the 107 minocycline-resistant isolates revealed that tetracycline resistance genes were identified in about one third, among which the *tet*(B) gene was present in 41.12% (n = 44/107) followed by the *tet*(A) gene in 22.43%

Table 1. PCR Primer Pairs Used in the Study

Resistance gene	*PCR primer sequence 5′–3′*	*Amplicon size (bp)*
$_{bla}$NDM	CACCT CATGTTTGAATTCGCC CTCTGTCACATTCGAAATCGC	984
$_{bla}$OXA 48	TATATTGCATTAAGCAAGGG CACACAAATACGCGCTAACC	848
*tet*A	GCT ACA TCC TGC TTG CCT TC CAT AGA TCG CCG TGA AGA GG	210
*tet*B	TTG GTT AGG GGC AAG TTT TG GTA ATG GGC CAA TAA CAC CG	659
*tet*K	TCG ATA GGA ACA GCA GTA CAG CAG ATC CTA CTC CTT	169
*tet*M	GTG GAC AAA GGT ACA ACG AG CGG TAA AGT TCG TCA CAC AC	406
*tet*S	CAT AGA CAA GCC GTT GAC C ATG TTT TTG GAA CGC CAG AG	667

($n = 24/107$), while *tet(K), tet*(M), and *tet*(S) genes were absent in all the isolates (Table 2).

Screening of β-lactamses revealed that 27.10% (n = 29/107) isolates showed the presence of bla_{NDM} gene, while $bla_{\mathrm{OXA\text{-}48}}$ gene was detected in 22.43% ($n = 24/107$) isolates (Table 2). It was worth noting that 24.3% ($n = 26/107$) of these isolates showed the presence of resistant genes from both groups (β-lactam and tetracycline).

The phenotypic MICs of each agent against all carbapenem nonsusceptible isolates are summarized in Table 3.

Raised MIC levels for minocycline were observed in our study, wherein 5 (4.76%) of our clinical Gram-negative isolates were inhibited at ≤8 mg/L and 28 (15.23%) were inhibited at ≤16 mg/L. Omadacycline was able to inhibit 14 (13.08%) of the isolates at ≤4 mg/L (susceptible breakpoint for *Enterobacterales*). It was worth noting that omadacycline was significantly more active against *Citrobacter spp.* (100%), *Klebsiella pneumoniae* (18.52%), and *E. coli* (12.5%) as compared to *P. aeruginosa* (9.68%) and *Acinetobacter baumannii* (7.14%) at the breakpoint concentration. Based on the cutoff value proposed, 15.09% (n = 16/106) isolates resistant to minocycline were inhibited by omadacycline.

Discussion

Enterobacterales group of bacteria and *Acinetobacter* species are common human pathogens implicated in various critical infections and are associated with high morbidity and mortality. This is attributed to the emergence and spread of antibiotic resistance. Carbapenems, which were frequently used in the past in critical care units to counter multidrug-resistant infections, are rapidly losing their vigour. The emergence of CRGNB is presently an eminent global health challenge.[22] As the burden is disproportionately high, clinicians are left with limited safe and effective therapeutic choices. Minocycline, a tetracycline derivative, is being frequently used as a carbapenem-sparing antimicrobial. However, the efficacy of minocycline to overcome carbapenem resistance has decreased with time. In the present study, 15.29% of the carbapenem-resistant clinical isolates were resistant to minocycline. A study from Brazil noted minocycline resistance of about 40% among multidrug-resistant GNBs.[23] Antimicrobial sensitivity profiling of the isolates recovered from central line-associated bloodstream infection in Northern India documented 50% minocycline resistance among GNBs.[24] Similar pattern (50% susceptibility toward minocycline) was observed in an Indian study.[25] Tigecycline Evaluation and Surveillance Trial, which is a global antimicrobial susceptibility surveillance study, ongoing since 2004, reported a global minocycline resistance rate of 15.5% for *A. baumannii* isolates and 28.6% for *K. pneumoniae* isolates.[26] A study performed in Chennai, India, showed that less than 70% of their MDR isolates were susceptible to minocycline.[27] Another report from South India demonstrated low susceptibility to minocycline

Table 2. Distribution of NDM, OXA, and Tetracycline Genes Among the Various Microorganisms Recovered from the Clinical Samples

S.N.	*Organism*	*No. of isolates (%)*	*Carbapenem gene* bla$_{NDM}$	bla$_{OXA\text{-}48}$	*Tetracycline gene* *Tet(A)*	*Tet(B)*
1	*Pseudomonas aeruginosa*	n = 31/107 (28.97%)	8 (25.80%)	5 (16.13%)	8 (25.80%)	12 (38.71%)
2	*Klebsiella pneumoniae*	n = 27/107 (25.23%)	5 (18.52%)	9 (33.33%)	4 (14.81%)	9 (33.33%)
3	*Escherichia coli*	n = 16/107 (14.95%)	3 (18.75%)	5 (31.25%)	5 (31.25%)	9 (56.25%)
4	*Acinetobacter spp.*	n = 14/107 (13.08%)	6 (42.85%)	1 (7.14%)	3 (14.29%)	7 (50.00%)
5	*Enterobacter spp.*	n = 7/107 (6.5%)	4 (57.14%)	1 (14.29%)	1 (14.29%)	2 (28.57%)
6	*Citrobacter spp.*	n = 6/107 (5.6%)	2 (33.33%)	1 (16.67%)	2 (33.33%)	3 (50.00%)
7	*Morganella spp.*	n = 3/107 (2.8%)	1 (33.33%)	2 (66.67%)	1 (33.33%)	1 (33.33%)
8	*Proteus spp.*	n = 3/107 (2.8%)	0 (00.00%)	0 (00.00%)	0 (00.00%)	1 (33.33%)
	Total	107	29 (27.10%)	24 (22.43%)	24 (22.43%)	44 (41.12%)

Table 3. Distribution of Minimum Inhibitory Concentration Values for Individual Pathogens

Organism/No. of isolates	*Antibiotics (range tested in μg/mL)*	*CLSI/European Committee on Antimicrobial Susceptibility Testing (EUCAST) breakpoint ≤S/≥R*	*Number of isolates with MIC (μg/mL)*									
			≤1	*≤2*	*4*	*8*	*16*	*32*	*64*	*128*	*≥256*	*% R*
Pseudomonas (31)	Imipenem (0.25–256)	≤1/≥4			2	3	1	3	4	7	11	100
	Meropenem (0.25–256)	≤1/≥4			4	1	4	2	3	8	9	100
	Minocycline (0.25–0.25–256)	≤4/≥16					9	1	7	11	3	100
	Omadacycline (0.25–256)				3	1	2	3	9	5	8	
Klebsiella pneumoniae (27)	Imipenem (0.25–256)	≤1/≥4	—	—	—	2	4	6	5	3	7	100
	Meropenem (0.25–256)	≤1/≥4	—	—	3	5	2	6	3	4	4	100
	Minocycline (0.25–0.25–256)	≤4/≥16			1	3	3	1	9	4	6	88.8
	Omadacycline (0.25–256)		—		5	6	8	2	4	1	1	
Escherichia coli (16)	Imipenem (0.25–256)	≤1/≥4			2	1	—	1	1	3	8	100
	Meropenem (0.25–256)	≤1/≥4					2	1	4	3	6	100
	Minocycline (0.25–0.25–256)	≤4/≥16					8	1	2	0	5	100
	Omadacycline (0.25–256)				2	5	3	1	2	1	2	
Acinetobacter (14)	Imipenem (0.25–256)	≤1/≥4	—					2	1	5	6	100
	Meropenem (0.25–256)	≤1/≥4	—				1	1	2	5	5	100
	Minocycline (0.25–256)	≤4/≥16				1	4	0	1	2	6	100
	Omadacycline (0.25–256)		—		1	5	1	2	1	1	3	
Enterobacter (7)	Imipenem (0.25–256)	≤1/≥4	—						1	2	4	100
	Meropenem (0.25–256)	≤1/≥4	—						3	1	3	100
	Minocycline (0.25–0.25–256)	≤4/≥16					1	3	0	1	2	100
	Omadacycline (0.25–256)		—			4	3					
Citrobacter (6)	Imipenem (0.25–256)	≤1/≥4	—	—	—	2	1	3				100
	Meropenem (0.25–256)	≤1/≥4			2	1	2	1				100
	Minocycline (0.25–0.25–256)	≤4/≥16				2	2	2				66.7
	Omadacycline (0.25–256)		2	3	1							
Proteus (3)	Imipenem (0.25–256)	≤1/≥4							1	2		100
	Meropenem (0.25–256)	≤1/≥4						1	1	1		100
	Minocycline (0.25–0.25–256)	≤4/≥16					1	1	1			100
	Omadacycline (0.25–256)						2	1				
Morganella (3)	Imipenem (0.25–256)	≤1/≥4								3		100
	Meropenem (0.25–256)	≤1/≥4						1	2			100
	Minocycline (0.25–0.25–256)	≤4/≥16						2	1			100
	Omadacycline (0.25–256)							2	1			

for *Klebsiella* (36.5%) and *Acinetobacter* isolates (50%).[28] Variable sensitivity patterns may be related to the regional practices, the organisms studied, patient profile, and the type of reporting centers (higher resistance at tertiary care centers).

This study also aimed to identify the diversity and distribution of tetracycline-resistant genes among carbapenem non-susceptible GNB bacteria. Genetic analysis revealed the presence of two different *tet* genes (*tet*A and *tet*B) among 107 clinically significant bacterial isolates. *Tet* genes are responsible for imparting resistance against the tetracycline group of antibiotics. *Tet* genes were most abundantly found in *Pseudomonas* strains, which accounted for almost 29% of all isolates. The co-existence of bla_{NDM} and bla_{OXA}-48 genes with *tet* genes among the majority of the isolates indicates a grim situation where clinicians are left with very limited options to treat infections caused by resistant GNBs.

Pseudomonas and *Klebsiella spp*. infections, two frequent but hard-to-treat causes for critical care admissions, are also known to carry *tet* gene. It is thought to be an important factor for the development of resistance. A multi-centric study by Hoban et al. documented minocycline resistance in 29.6% isolates of *K. pneumoniae* and the rates increased to 47.8% among the isolates resistant to carbapenem as well.[26,29] In a cross-sectional study from Japan (2017), the prevalence of carbapenem-resistant *Enterobacterales* among patients recuperating at home after prolonged ICU (intensive care unit) stay was 12.2%. Most common species were *E. coli* (57.5%) and *K. pneumoniae* (39.1%), more than 95% being carbapenemase-producers.[27] Monitoring of cross-resistance between tigecycline and minocycline resistance patterns for GNBs is recommended to keep a check on the utilization of this drug in the health sector.[30] Phenotypic and molecular characterization of the resistant clinical isolates is important to provide information on the epidemiological characteristics of such pathogens. The *tet*(A) gene generally confers resistance to tetracycline alone, whereas *tet*(B) confers resistance to minocycline as well. However, a study on multiresistant *A. baumannii* strains demonstrated the presence of both *tet*(A) and *tet*(B) type of genes in minocycline-resistant clinical isolates.[10]

Another drug approved by the FDA for the treatment of drug-resistant bacterial infections is omadacycline. In the present study, 11.22% of clinical isolates were inhibited at ≤4 mg/L (breakpoint), which is much lower than that documented in the SENTRY antimicrobial surveillance program.[31]

Activity of omadacycline against GNBs was not as effective as documented in studies from China and America.[14,32,33] However, it was worth noting that a few highly resistant bacterial isolates included in our study tested sensitive to omadacycline. Higher MIC among the XDR (Extensively drug-resistant) isolates suggest preexisting tolerance to omadacycline. This highlights the possibility of similar molecular mechanisms evading drug action in resistant bacteria. Moreover, the patient cohort in the present study was referred from other hospitals, and already exposed to multiple antibiotics before admission, which might be the reason for such a high antimicrobial resistance pattern.

Recently, identification of tetracycline-resistant genes among GNBs has been reported from several countries. This is the first report from India on the distribution of tetracycline-resistant genes (*tet*A and *tet*B) among carbapenem-resistant clinical isolates and evaluation of the activity of both the broad-spectrum last-line drugs, that is, minocycline and omadacycline against these drug-resistant bugs. We focused on CRGNB among intensive care patients, which are clinically the most important and difficult-to-treat organisms. The study was adequately powered. The MIC levels of the second-line and last-resort antimicrobial agents were elaborated among the study samples, which can prove useful in developing the treatment protocol. Our study has a few limitations as well. Being a tertiary care referral center and samples being taken from patients already exposed to multiple antibiotics, the prevalence of bacterial resistance may not be a true representation of the pattern in the entire region. Further characterization in a multicenter study and correlation with the clinical response of patients is warranted.

Conclusion

The results of the present study document a significant resistance to minocycline and omadacycline among CRGNB. The resistant strains carried *tet*B and *tet*A genes in nearly half of the isolates. Minocycline should be judiciously used as a cost-effective alternative to colistin and tigecycline in critical care units. Determination of MIC values and *tet* gene status (*tet*B and *tet*A) may be helpful before clinical use. The *in vitro* effect of omadacycline in the present study among isolates from infections other than that indicated in the literature provides references for other clinical uses as well and needs further evaluation before its clinical use in ICU patients.

Authors' Contributions

R.S.: Conceptualization, project administration, supervision, validation, and writing—original draft, review, and editing. J.J.: Data curation, investigation, resources, and writing—original draft. A.P.: Formal analysis, validation, and writing–review and editing. C.S.: Conceptualization, project administration, supervision, resources, funding acquisition, and project administration. P.M.: Formal analysis, software, and validation. R.S.K.M.: Supervision and writing—review and editing. U.G.: Resources, supervision, and writing—review and editing.

Disclosure Statement

No competing financial interests exist.

Funding Information

The present study was approved by the institutional ethical committee of SGPGIMS (internal funding for an intramural project)—Grant no. PGI/DIR/RC/916/2021.

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Address correspondence to:
Chinmoy Sahu, MD
Sanjay Gandhi Post Graduate Institute of Medical Sciences
SGPGI Rae Bareilly Road
Type IV/7
Lucknow 226014
Uttar Pradesh
India

E-mail: sahu.chinmoy@gmail.com